1.4 Formulae

1.4.1 Symbolic Computation

One of the most important things about *Mathematica* is that it can do *symbolic*, as well as *numerical* calculations. This means that *Mathematica* can work with algebraic formulae as well as with numbers.

This assigns p to have the numerical value 2.

```
In[1]:= p = 2
Out[1]= 2
```

Using the numerical value for p, *Mathematica* can work out the numerical value of this expression.

```
In[2]:= 3p - p + 2
Out[2]= 6
```

q has no numerical value in this case. *Mathematica* can nevertheless simplify this expression *symbolically*.

```
In[3]:= 3q - q + 2
Out[3]= 2 + 2 q
```

numerical computation	`3 + 62 - 1` ⟶ `64`
symbolic computation	`3x - x + 2` ⟶ `2 + 2 x`

Numerical and symbolic computations.

You can type any algebraic expression into *Mathematica*. The expression is printed out in an approximation to standard mathematical notation.

```
In[6]:= -1 + 2x + x^3
                    3
Out[6]= -1 + 2 x + x
```

Mathematica automatically carries out standard algebraic simplifications. Here it combines x^2 and $-4x^2$ to get $-3x^2$.

```
In[7]:= x^2 + x - 4 x^2
                2
Out[7]= x - 3 x
```

You can type in any algebraic expression, using the operators listed on page 37. You can use spaces to denote multiplication. Be careful not to forget the space in `x y`. If you type in `xy` with no space, *Mathematica* will interpret this as a single symbol, with the name `xy`, not as a product of the two symbols `x` and `y`.

Mathematica rearranges and combines terms using the standard rules of algebra.

```
In[8]:= x y + 2 x^2 y + y^2 x^2 - 2 y x
                     2       2  2
Out[8]= -(x y) + 2 x  y + x  y
```

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Here is another algebraic expression.

```
In[9]:= (x + 2y + 1)(x - 2)^2
```

$$Out[9]= (-2 + x)^2 (1 + x + 2 y)$$

The function `Expand` multiplies out products and powers.

```
In[10]:= Expand[%]
```

$$Out[10]= 4 - 3 x^2 + x^3 + 8 y - 8 x y + 2 x^2 y$$

`Factor` does essentially the inverse of `Expand`.

```
In[11]:= Factor[%]
```

$$Out[11]= (-2 + x)^2 (1 + x + 2 y)$$

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